

GPT: Decoder-Only Transformer

Deep Learning

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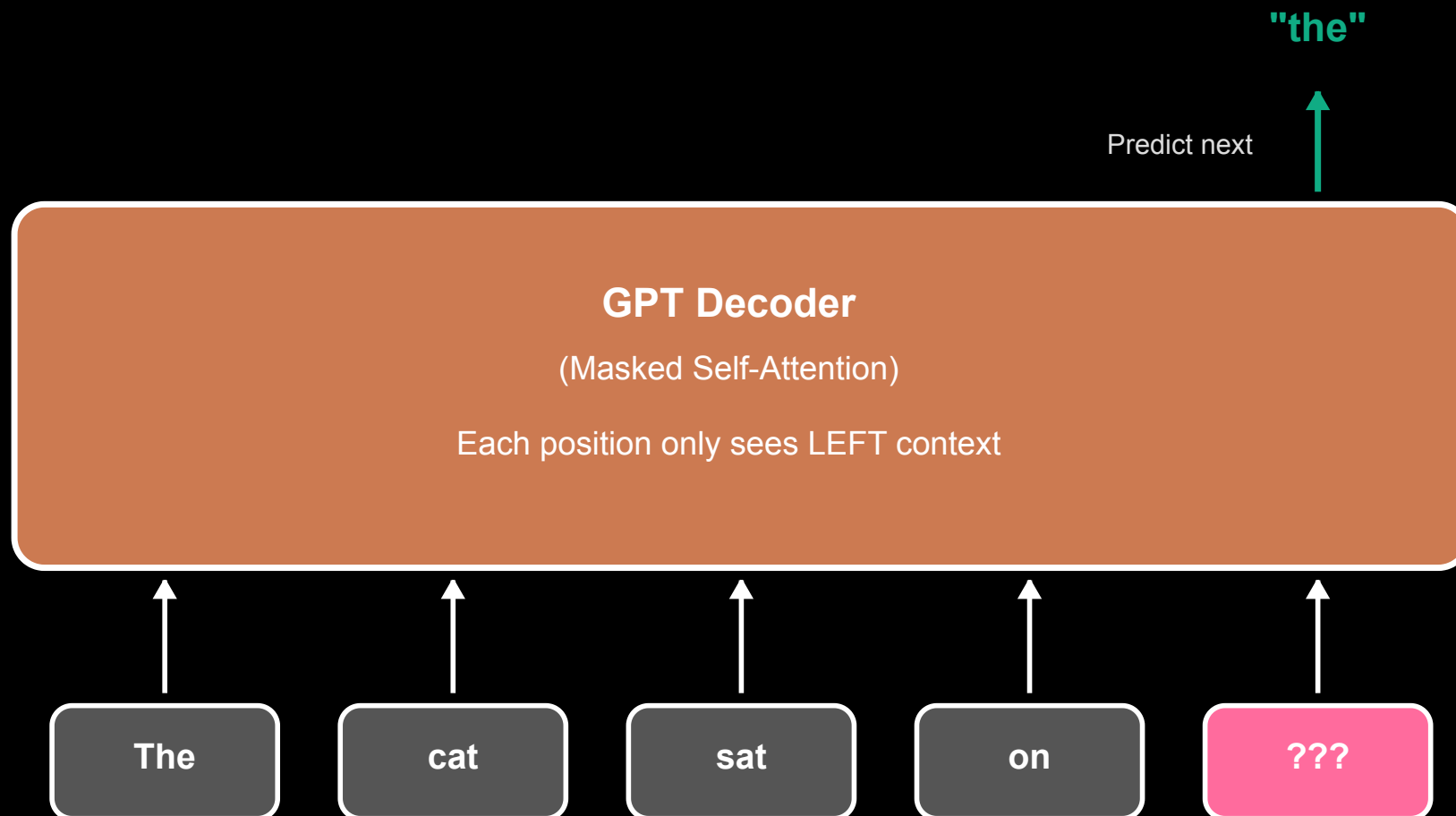
Contents of This Video

In this video, we will cover:

- GPT: decoder-only architecture
- Causal (left-to-right) attention
- Autoregressive language modeling
- Scaling laws: bigger is better
- From GPT to ChatGPT

GPT: Decoder-Only

GPT: Decoder-Only, Left-to-Right



Autoregressive Generation

Training:

Predict next token given previous tokens:

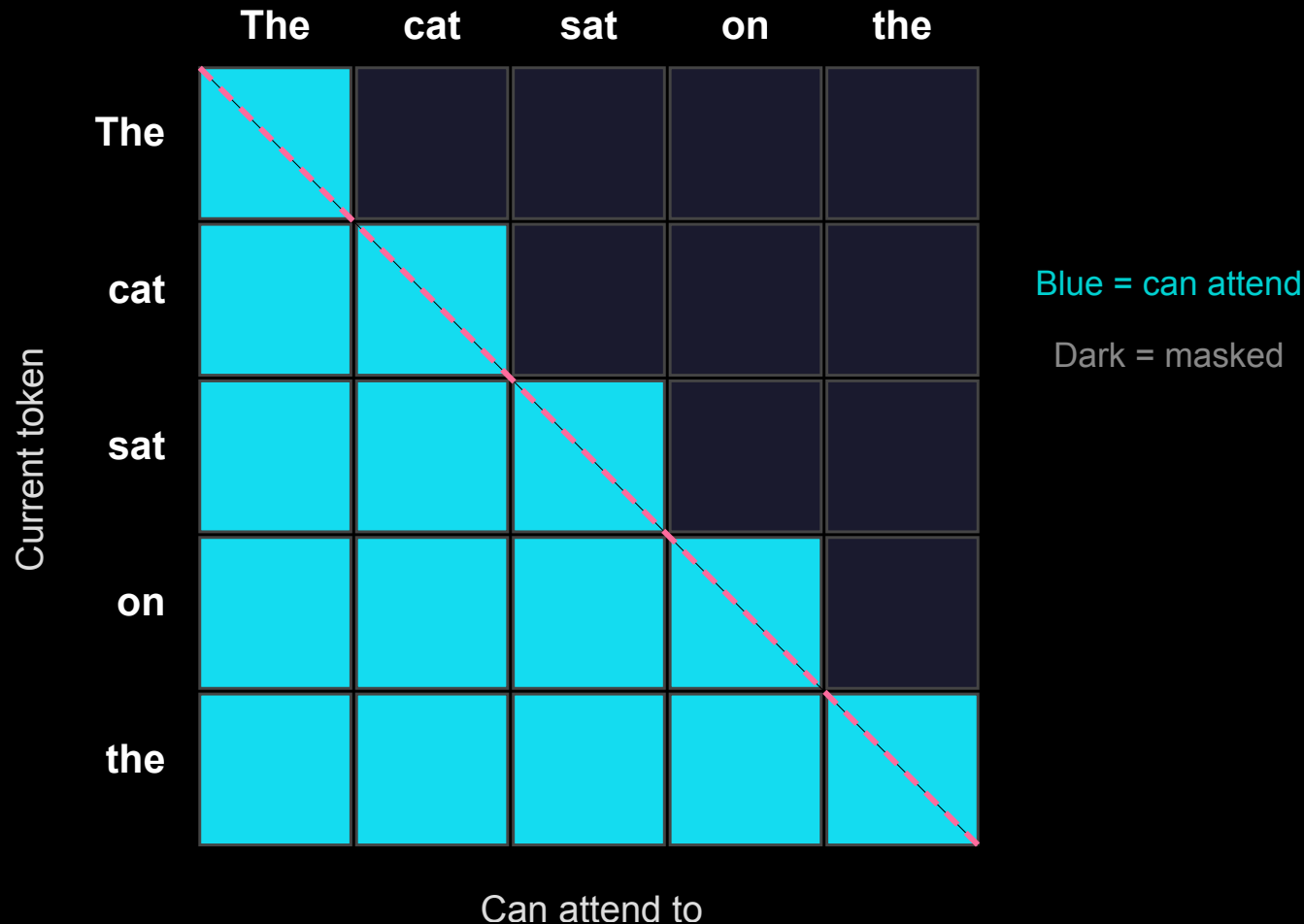
$$P(\text{token}_t | \text{token}_1, \dots, \text{token}_{t-1})$$

Generation:

1. Start with prompt: “The cat”
2. Predict next: “sat” (sample from distribution)
3. Append: “The cat sat”
4. Predict next: “on”
5. Continue until done...

Causal (Masked) Attention

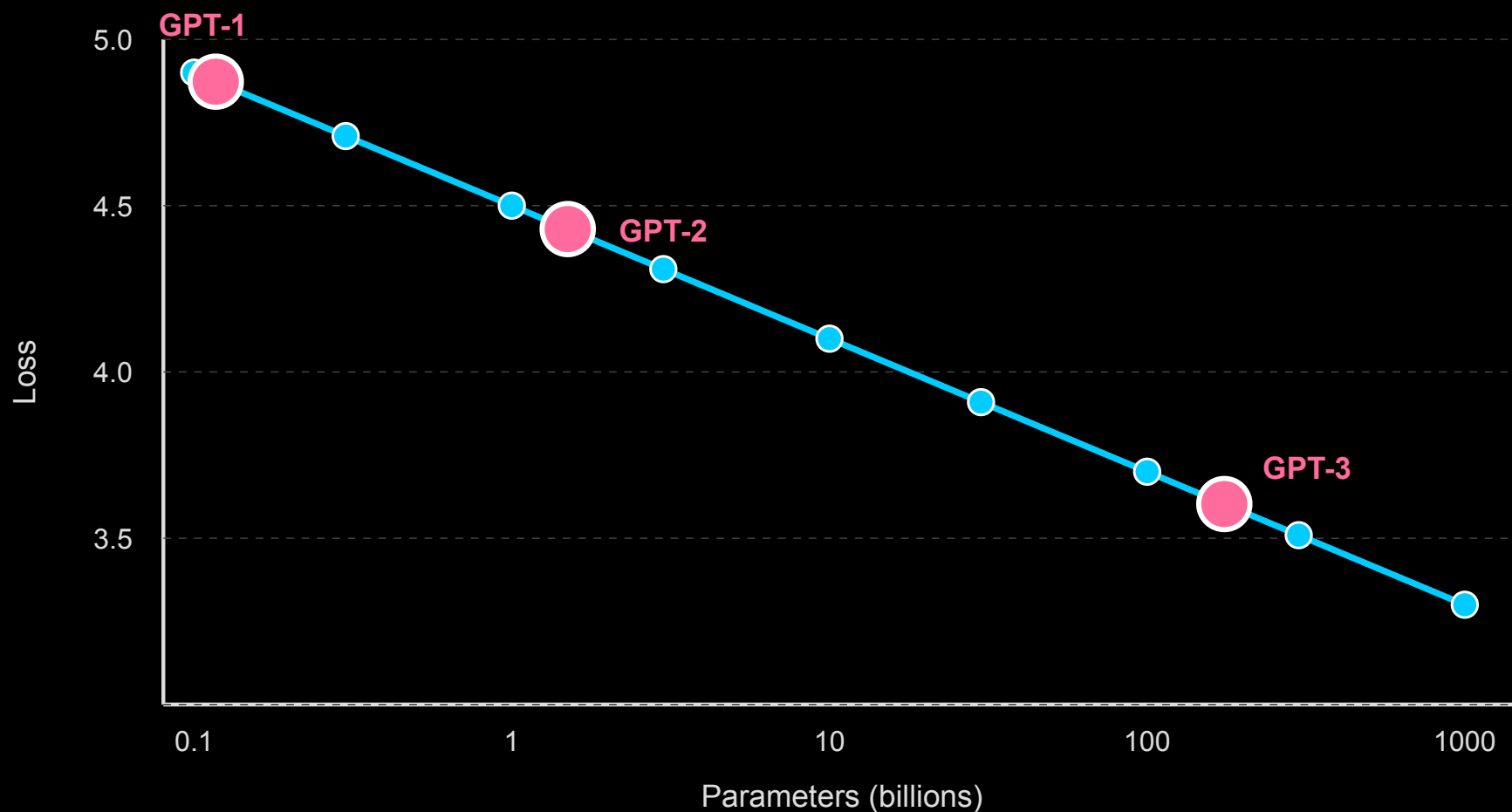
Causal Attention: Only See Past



Blue = can attend; white = masked (can't see)

Scaling Laws

Scaling Laws: Bigger Models = Lower Loss



Loss decreases predictably with scale (parameters, data, compute)

The GPT Family

Model	Year	Parameters	Training
GPT-1	2018	117M	Books
GPT-2	2019	1.5B	WebText
GPT-3	2020	175B	Massive web
GPT-4	2023	~1T?	Unknown

GPT-3 showed “emergent abilities”

ChatGPT = GPT-3.5 + RLHF (reinforcement learning from human feedback)

What We've Covered

In this video, we've learned:

- GPT: decoder-only Transformer (generation, not understanding)
- Causal attention: only see previous tokens
- Autoregressive: predict next token, append, repeat
- Scaling laws: more parameters = better performance
- GPT family: 117M $\rightarrow$ 175B+ parameters
- ChatGPT = GPT + RLHF for helpfulness